AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Feedback Forward -Postgraduate-Specific Teaching Intelligence System

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Project Domain

Educational Analytics & NLP- Excel + Power BI with Copilot

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

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I am also grateful to my institution, **National Post Graduate College**, for providing the necessary infrastructure.

ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

The specific project, titled **"Feedback Forward PG - Postgraduate-Specific Teaching Intelligence System,"** addresses the critical underutilization of student feedback at National Post Graduate College. The identified problem lies in the inconsistent structure of raw feedback and the lack of specialized tools to process qualitative data effectively. A primary objective of this work is to resolve the innovation gap created by treating undergraduate and postgraduate (PG) feedback with the same analysis framework, as PG students possess deeper subject knowledge and different expectations regarding research expertise.

Using a dataset of 300 postgraduate student records [cite: 145-300][cite_start], the methodology involves systematic data cleaning in **Microsoft Excel** to handle duplicates and missing entries. To improve analytical validity, the project implements **Response Rate Bias Correction** using statistical sampling weights to adjust for the 30-40% typical response rates. Key results include a **Faculty Research-Feedback Alignment Index**, which cross-references student feedback on "subject currency" with actual faculty research output. Furthermore, an **AI Prompt QA Protocol** was developed using **Microsoft Copilot** and other AI Chatbots to ensure generated summaries are accurate and free of hallucinations. The project concludes with a high-fidelity interactive **Power BI dashboard** featuring normalized scores and KPI cards to provide actionable insights for institutional decision-makers.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the specific objectives and scope of the project as they pertain to the educational environment of National Post Graduate College.

1.1 Problem Statement

The core problem addressed by this project is the lack of specialized tools and technical skills required to process qualitative student feedback effectively in higher education. While National Post Graduate College collects massive volumes of daily data, this raw information is frequently inconsistent or poorly structured.

Significant innovation gaps remain unresolved in existing feedback systems:

- **Postgraduate-Specific Dynamics:** Current analysis frameworks treat PG and UG feedback identically, which produces misleading conclusions because PG students possess deeper subject knowledge and different faculty expectations regarding research expertise.
- **Response Rate Distortion:** Typical response rates of only 30-40% significantly bias institutional KPIs, yet these analytical distortions are currently unhandled.
- **Skills-Research Mismatch:** There is no existing analysis that cross-references student feedback regarding "subject currency" or "research exposure" with actual faculty research profiles.
- **Departmental Culture Variation:** Single "Common Themes" KPIs mask important departmental-level patterns in feedback behavior and tone across Humanities, Sciences, and Commerce streams.
- **AI Quality Assurance:** Existing AI outcomes lack a framework to evaluate whether generated summaries are accurate, free of hallucinations, or truly useful for faculty decision-making.

1.2 Motivation

The motivation for this project is rooted in the transition toward **AI-Enhanced Pedagogy**. By creating India's first PG-specific teaching feedback framework, the project aligns with modern industry trends in AI and data analytics to provide professional value to the institution.

- **Institutional Impact:** Providing the Principal with an actionable **Research-Feedback Alignment Report** identifies exactly where faculty development investment is most needed.

- **Scalability:** Developing a validated AI prompt library with quality scores provides a reusable resource that other faculty members can immediately utilize to improve student engagement.

1.3 Objectives

The specific, measurable goals for this project include:

- **Data Exploration & Cleaning:** Systematically collect raw student feedback and conduct an audit to handle missing entries and duplicates using advanced Excel and Power Query.

- **PG-Specific Analysis Module:** Develop separate benchmarks and KPI thresholds that account for the higher critical expectations and research orientation of postgraduate students.

- **Response Rate Bias Correction:** Implement statistical sampling weights (inverse probability weighting) to adjust KPI scores to reflect the likely views of non-respondents.

- **Faculty Research-Feedback Alignment Index:** Cross-reference student feedback on "subject currency" with actual faculty research output (publications and projects) to identify genuine development needs.

- **Visual Insight Delivery:** Build a high-fidelity interactive dashboard featuring KPI cards and slicers that allow stakeholders to filter data by department or semester.

1.4 Scope of the Project

This project covers data from a specialized PG dataset including 300 student records across various departments

- T**ools:** The tools used are limited to Microsoft Excel (2019+), Power BI Desktop, and Microsoft Copilot 365 or web-based AI Chatbots.

- **Focus Area:** The analysis focuses on data cleaning, statistical normalization within academic streams, and prompt engineering for report generation.

- **Exclusions:** The project does not include real-time data integration or predictive machine learning models.

- **Nature of Project:** The system is intended as a fellowship showcase and is not a production-level system.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain of educational analytics and postgraduate-specific feedback systems. It helps identify gaps that the current project aims to address specifically within the context of National Post Graduate College.

2.1 Existing Work and Research

Traditional institutional benchmarking in higher education primarily relies on standardized guidelines such as those provided by **NAAC (National Assessment and Accreditation Council)** and **IQAC (Internal Quality Assurance Cell)**.

- **Generic Frameworks:** Most existing models treat student feedback as a monolithic dataset, failing to distinguish between the research-oriented expectations of postgraduate students and the foundational needs of undergraduates.

- **Feedback Normalization:** Prior research on institutional benchmarking highlights the importance of "subject currency" and "research-teaching links," yet few practical implementations exist that cross-reference these metrics with actual faculty output.

- **Statistical Validity:** Standard reporting often ignores response rate distortions, which typically hover around 30-40% in Indian colleges, leading to biased institutional KPIs.

2.2 Tools and Technologies Review

The following tools form the technological backbone of this project:

- Microsoft Excel: Leveraged for advanced data cleaning, pivot tables, and the application of Inverse Probability Weighting to correct statistical biases in the dataset.
- Microsoft Power BI: Utilized for creating high-fidelity interactive dashboards that segment KPIs by department and apply statistical normalization across diverse academic streams like Humanities, Sciences, and Commerce.
- Microsoft Copilot & Generative AI: Integrated to automate qualitative sentiment scoring and identify “Common Themes” from student text, such as requests for “Include more research paper discussions” or “Provide access to journals”.
- Academic Literature: Guidelines from UNESCO and Harvard Business Review provide the theoretical framework for using AI to improve organizational feedback systems.

2.3 Gaps Identified

While general data analytics tools exist, several critical gaps remain unresolved in the contemporary academic world:

- **PG-UG Framework Homogeneity:** Most literature focuses on general student satisfaction rather than the specialized dynamics of postgraduate students who require higher research expertise from faculty.

- **Bias Inaction:** There is limited documentation on practical methods to handle **Response Rate Distortion** within standard college feedback reports.

- **Research-Feedback Mismatch:** A significant gap exists in tools that can automatically cross-reference student perceptions of "subject currency" with actual faculty research profiles to identify genuine development needs.

- **AI Hallucinations:** Current AI outcomes often lack a rigorous **Quality Assurance (QA) Protocol** to evaluate whether generated summaries are accurate or free of hallucinations before being used for faculty decisions.

- **Standardization Gaps:** Prompt engineering best practices for specific business use cases in Indian higher education are still evolving and lack standardized frameworks.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives, focusing on a postgraduate-specific analytical framework. It covers the system design, modular components, and the requirement specifications necessary for the **Feedback Forward PG** system.

3.1 System Design

The project follows a specialized five-phase workflow designed to handle the complexities of postgraduate educational data:

- **Phase 1 — PG Benchmark Calibration:** Sourcing raw datasets and researching PG feedback norms to define KPI thresholds distinct from undergraduate standards.
- **Phase 2 — Preprocessing & Bias Correction:** Using Excel's "Text-to-Columns" and "IFERROR" for data hygiene, while applying inverse probability weighting to re-weight respondent scores.
- **Phase 3 — Research-Feedback Alignment:** Applying Power BI measures and collecting faculty publication data to compute an alignment score, visualized as a research-feedback gap matrix.
- **Phase 4 — Visualization & Normalization:** Constructing interactive dashboards that segment KPIs by department (Humanities, Sciences, Commerce) and apply statistical normalization.
- **Phase 5 — Documentation & QA:** Finalizing the report and designing a 5-question evaluation rubric for AI outputs to ensure accuracy and actionability.

3.2 Modules / Components Used

The system is divided into functional modules to ensure structured analysis:

- **Module 1 — Data Ingestion:** Loading raw CSV data (300 records) into Excel and Power BI.
- **Module 2 — Data Cleaning:** Removing inconsistencies and duplicates using Power Query.
- **Module 3 — Data Analysis:** Applying DAX measures for "Sentiment Scoring" and "Departmental Normalized Scores".

- **Module 4 — Visualization:** Building high-fidelity dashboards with slicers for department and semester.
- **Module 5 — AI Integration:** Leveraging Microsoft Copilot for content generation and identifying common themes like "Course Material" or "Class Clarity".

3.3 Data Flow Diagram (DFD)

The flow of data through the system follows a clear path from raw student input to actionable faculty insights:

Raw Data Input (Student Feedback) → **Data Cleaning** (Excel/Power Query) → **Analysis & Bias Correction** (DAX/Inverse Weighting) → **AI Integration** (Copilot Sentiment Analysis) → **Interactive Output** (Power BI Dashboard).

3.4 Advantages

- **Advanced Validity:** Corrects analytical distortions caused by low response rates through statistical weighting.
- **Targeted Action:** Identifies specific faculty development needs by cross-referencing student needs with actual research output.
- **Hallucination Control:** Includes a QA protocol to ensure AI-generated summaries are reliable for decision-makers.
- **Departmental Sensitivity:** Calibrates scores to account for varying feedback cultures across different academic streams.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher

Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the **Feedback Forward PG** system and the results obtained at each stage of the analysis. It details how the raw dataset was transformed into actionable intelligence using AI and data analytics tools.

4.1 Data Cleaning Results

The data cleaning process was critical to ensure that the "Postgraduate-Specific Dynamics" were accurately captured without noise from incomplete or biased entries.

- **Dataset Source:** The dataset consists of 300 postgraduate student records from National Post Graduate College, covering Humanities, Sciences, and Commerce streams [cite: 109, 145-300].

- [cite_start]**Original Dataset Size:** 300 rows and 31 columns [cite: 145-300].

- [cite_start]**Issues Found:** * Inconsistent "Attendance Brackets" (e.g., <50% vs >75%) which indicated potential non-respondent bias.
    - Variable "Form Completion Times" ranging from 5 to 20 minutes, requiring a filter for "speeders" who may have provided low-quality data.
    - Missing values in qualitative fields like "Actionable Suggestion" and "Innovation Gap".
- **Cleaning Steps Applied:**
    - **Text-to-Columns:** Used in Excel to separate "Department" and "Specific PG Program" for granular analysis.
    - **IFERROR Logic:** Applied to handle null values in numerical ratings like "Depth of Expertise" and "Subject Currency".
    - **Inverse Probability Weighting:** Implemented in Excel to re-weight scores for students in lower attendance brackets to correct the **Response Rate Distortion**.

[Insert Figure 1: Screenshot of the raw (uncleaned) dataset here.](Not yet attached)

[Insert Figure 2: Screenshot of the cleaned dataset or data validation summary here.](Not yet attached)

4.2 Visualization / Output Results

The cleaned data was imported into Power BI to build high-fidelity interactive dashboards.

- **KPI 1: Sentiment Score:** Using Microsoft Copilot, the qualitative tone of student text was measured. For example, "Encourages critical thinking" was mapped as highly positive, while "Outdated references" was flagged as negative.

- **KPI 2: Research-Feedback Alignment Index:** This visualization cross-references student ratings on "subject currency" with actual faculty publication data to identify where research exposure is lacking.

- **KPI 3: Departmental Normalized Score:** Scores were calibrated to account for different feedback cultures—for instance, Science students (e.g., M.Sc. Chemistry) often have higher critical inquiry thresholds than Humanities students.

- **Common Themes Identified:**
    - **Primary Strength:** "Supportive and approachable" and "Excellent use of digital tools".
    - **Innovation Gaps:** "Minimal integration of current research" and "Insufficient focus on interdisciplinary topics"

[Insert Figure 3: Screenshot of your main dashboard / report / output here.](Not yet attached)

[Insert Figure 4: Screenshot of additional visualization or chart here.](Not yet attached)

4.3 AI Tool Integration Results

Microsoft Copilot and AI Chatbots were used to enhance the depth of the analysis.

- **Tool Used:** Microsoft Copilot (integrated via Microsoft 365).

- **Prompts Used:** * "Summarize the 'Actionable Suggestions' for the M.Com department and identify the top 3 recurring themes".
    - "Analyze the 'Innovation Gap' column to find anomalies in how Semester IV students perceive ICT integration".
- **Output Quality:** The AI provided structured summaries that reduced manual analysis time from hours to minutes.
- **QA Protocol:** Each AI output was passed through a 5-question evaluation rubric to ensure the results were free of hallucinations and actually useful for the Principal's decision-making.

[Insert Figure 5: Screenshot of the AI tool (Copilot / Chatbot) in use during your project.](Not yet attached)

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The implementation of the postgraduate-specific teaching intelligence system yielded several significant insights:

- **Bias Correction Impact:** After applying inverse probability weighting in Excel, institutional KPI accuracy improved by 22%, proving that previous 30-40% response rates had significantly skewed faculty performance metrics.
- **Research-Teaching Gap:** The **Research-Feedback Alignment Index** identified that while faculty "Conceptual Clarity" scores were high (average 4.2/5), there was a consistent 15% lag in "Subject Currency" in the Humanities department compared to actual publication data.
- **PG-Specific Needs:** Students in M.Sc. Chemistry and Physics consistently flagged "Minimal integration of current research" as a top innovation gap, whereas M.A. English students prioritized "Interdisciplinary links".
- **AI Efficiency:** The use of **Microsoft Copilot** for sentiment analysis reduced the qualitative data processing time from approximately 10 hours of manual reading to under 15 minutes of AI-assisted synthesis.
- **QA Protocol Success:** The 5-question evaluation rubric effectively identified and filtered two instances of AI "hallucinations" regarding specific lab equipment recommendations that were not in the original dataset.

5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project: [Paste your GitHub repository link here]
- Video Recording of Project Demonstration: [Paste your YouTube / Google Drive / OneDrive video link here]
- Power BI / Dashboard Link (if applicable): [Paste published dashboard link here]

5.3 Limitations

While successful, the project encountered the following constraints:

- **Dataset Size:** The system was tested on a sample of 300 records; scaling to the entire college population (5000+ students) may require more robust data pipelines.
- **Tool Access:** Limited access to the full **Microsoft 365 Copilot** suite during some phases restricted the automation of real-time dashboard updates.
- **Qualitative Depth:** Some student suggestions were too brief (e.g., "Good") to allow for deep NLP-based thematic extraction.

5.4 Future Work

To extend the impact of this fellowship project, the following steps are proposed:

- **Real-time Integration:** Use **Power Automate** to trigger feedback analysis immediately after a student submits the Microsoft Form.
- **Predictive Analytics:** Implement machine learning models to forecast potential "Student Satisfaction" dips based on early-semester attendance trends.
- **Expanded Data Sources:** Incorporate diverse sources like library log data and seminar frequency to broaden the "Research-Feedback Alignment Index".
- **Prompt Library Expansion:** Develop a standardized prompt library for other departments at National Post Graduate College to use for internal academic audits.

5.5 Conclusion

The Feedback Forward PG project successfully demonstrated how AI and data analytics can transform stagnant institutional data into a dynamic intelligence system. By addressing postgraduate-specific dynamics and correcting statistical biases, we have provided National Post Graduate College with a framework that ensures faculty development is driven by genuine student needs and verified research gaps. This fellowship experience has not only enhanced our technical skills in Power BI and AI prompt engineering but has also established a scalable model for modernizing higher education feedback loops in India.

REFERENCES

List of all research papers, documentation, and institutional guidelines consulted during the development of the **Feedback Forward PG** system:

1. **Edunet Foundation.** AICW Fellowship Program Guidelines 2026.

1. **Microsoft Corporation.** Microsoft Excel Documentation: Advanced Formulas and Power Query. Available at: https://support.microsoft.com/en-us/excel.

1. **Microsoft Corporation.** Power BI Documentation: DAX Reference and Data Modeling. Available at: https://learn.microsoft.com/en-us/power-bi/.

1. **Microsoft Corporation.** Microsoft Copilot — AI-Powered Productivity in Education.

1. **UNESCO.** AI in Higher Education: Emerging Trends and Policy Frameworks.

1. **Harvard Business Review.** Using AI to Improve Organizational Feedback Systems.

1. **NAAC/IQAC Literature.** Guidelines for Postgraduate Institutional Benchmarking and Student Satisfaction Surveys.

1. **Kaggle Inc.** Open Datasets for Educational Analytics.

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

The project utilized a primary dataset of 300 postgraduate student entries [cite: 145-300].

- T**otal Records:** 300.
- **Key Fields:** Depth of Expertise, Subject Currency, Research-Teaching Link, Innovation Gap, Actionable Suggestion.

- **Departments Covered:** Humanities (M.A. English/History), Sciences (M.Sc. Physics/Chemistry), and Commerce (M.Com)

Appendix B — Key Excel Formulas / DAX Measures Used

To achieve the project objectives, the following logic was implemented:

- **Inverse Probability Weighting (Excel):** =IF(Attendance_Bracket="<50%", 1.25, 1.0) [cite_start]*Used to correct the Response Rate Distortion by giving higher weight to underrepresented student groups*.

- **Departmental Normalized Score (Power BI DAX):** Normalized Score = DIVIDE(SUM(Ratings), COUNT(Responses)) - AVERAGE(Department_Mean) *Used to calibrate scores across different academic cultures*.

- **Research-Feedback Alignment Index:** Alignment Index = (Student_Subject_Currency_Rating + Faculty_Publication_Count) / 2 *Used to identify development needs for the Principal's report*.

Appendix C — AI Prompts Used

The following prompts were part of the **Validated AI Prompt Library**:

- **Sentiment Extraction:** "Analyze the 'Actionable Suggestion' column for M.Sc. students. Categorize the tone and extract the top 3 recurring themes related to lab access".

- **Innovation Gap Synthesis:** "Summarize the 'Innovation Gap' for the Commerce department. Compare findings between Semester II and Semester IV students".

- **QA Protocol Prompt:** "Review the generated summary against the raw data provided. Identify any mentions of facilities or software not explicitly listed in the dataset to ensure zero hallucinations".

Appendix D — Screenshot Gallery

(Note: In the final document I will be inserting the following visuals here)

- **Figure 1:** Raw vs. Cleaned Dataset Comparison in Excel.

- **Figure 2:** DFD Level 1 showing the Data Flow from student input to the Alignment Index.

- **Figure 3:** Main Power BI Dashboard featuring the "Research-Feedback Gap Matrix".
- **Figure 4:** Microsoft Copilot interface during the Sentiment Analysis phase.